

# COL362/632 Introduction to Database Management Systems

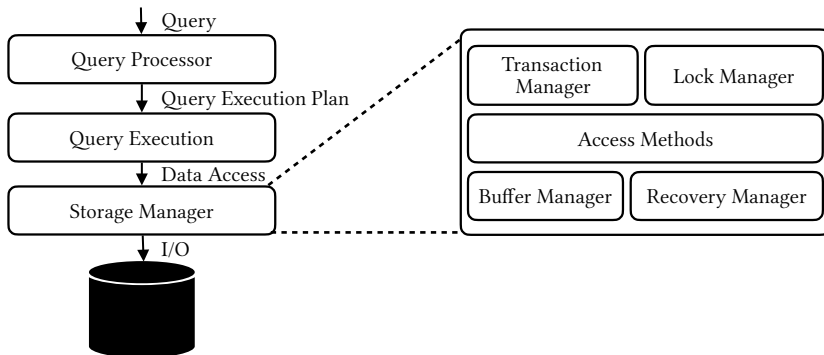
## Database Systems – Buffer Manager

Kaustubh Beedkar

Department of Computer Science and Engineering  
Indian Institute of Technology Delhi



# Database Storage

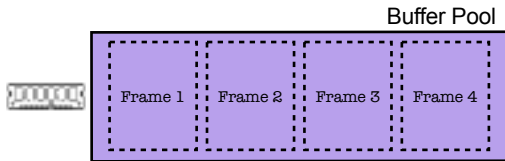


- ▶ DBMS stores database in files on disk
- ▶ Buffer pool manager takes care of memory management and Disk  $\xleftrightarrow{\text{Data}}$  Memory

# Buffer Manager

## Buffer Pool

- ▶ An **array** of fixed-size page in the memory
- ▶ Entry in the array is called a **frame**



## Buffer Manager

- ▶ Copies a page from (to) disk to (from) buffer pool



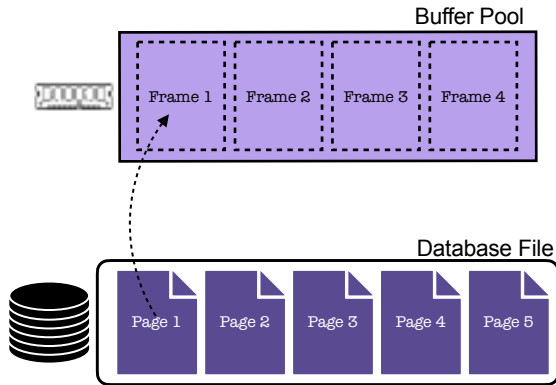
# Buffer Manager

## Buffer Pool

- ▶ An **array** of fixed-size page in the memory
- ▶ Entry in the array is called a **frame**

## Buffer Manager

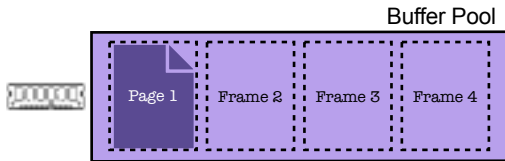
- ▶ Copies a page from (to) disk to (from) buffer pool



# Buffer Manager

## Buffer Pool

- ▶ An **array** of fixed-size page in the memory
- ▶ Entry in the array is called a **frame**



## Buffer Manager

- ▶ Copies a page from (to) disk to (from) buffer pool



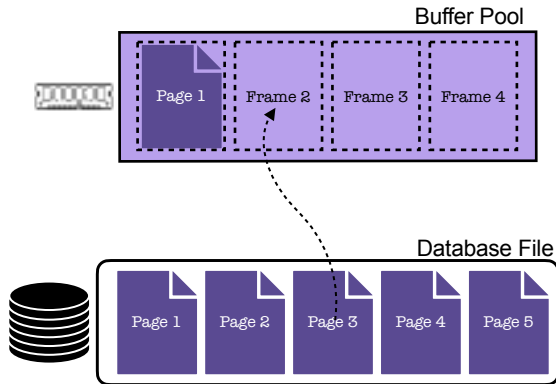
# Buffer Manager

## Buffer Pool

- ▶ An **array** of fixed-size page in the memory
- ▶ Entry in the array is called a **frame**

## Buffer Manager

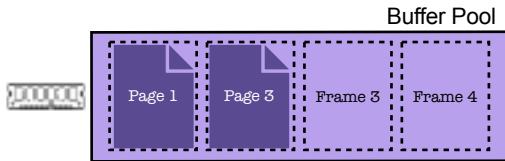
- ▶ Copies a page from (to) disk to (from) buffer pool



# Buffer Manager

## Buffer Pool

- ▶ An **array** of fixed-size page in the memory
- ▶ Entry in the array is called a **frame**



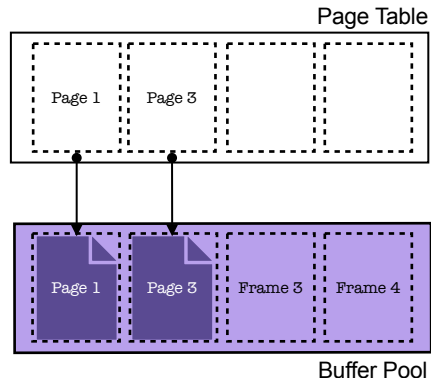
## Buffer Manager

- ▶ Copies a page from (to) disk to (from) buffer pool



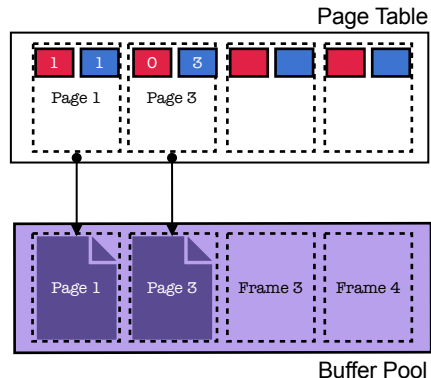
# Page Table

- ▶ In-memory data structure
- ▶ Tracks mappings of pages in memory (**note:** this is different from the page directory)
- ▶ Stores additional metadata per page
  - **dirty bit**
    - Whether or not the page has been modified in memory
  - **pin count**
    - Number of current queries using the page



# Page Table

- ▶ In-memory data structure
- ▶ Tracks mappings of pages in memory (**note:** this is different from the page directory)
- ▶ Stores additional metadata per page
  - **dirty bit**
    - Whether or not the page has been modified in memory
  - **pin count**
    - Number of current queries using the page



# Page Requests

- ▶ **Read (page)**

Read page from disk into buffer pool  
(if not already present)

- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

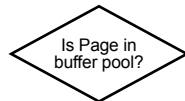
- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk

# Page Requests

- ▶ **Read (page)**

Read page from disk into buffer pool  
(if not already present)



- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk

# Page Requests

- ▶ **Read (page)**

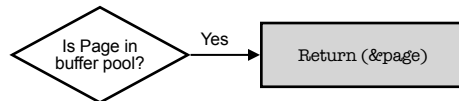
Read page from disk into buffer pool  
(if not already present)

- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk



# Page Requests

- ▶ **Read (page)**

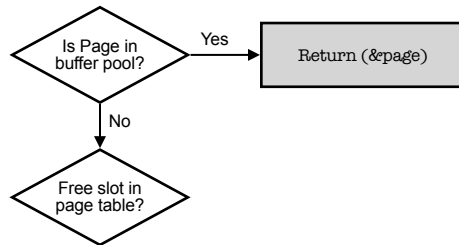
Read page from disk into buffer pool  
(if not already present)

- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk



# Page Requests

- ▶ **Read (page)**

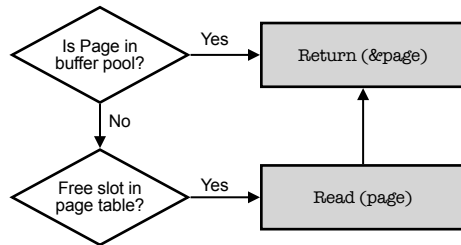
Read page from disk into buffer pool  
(if not already present)

- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk



# Page Requests

- ▶ **Read (page)**

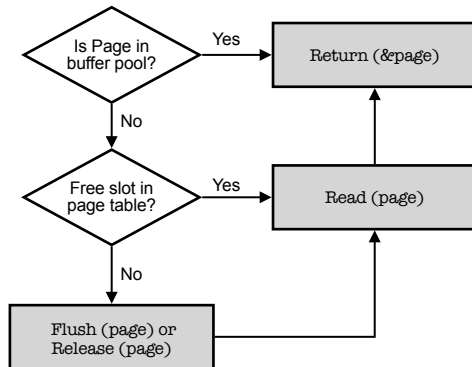
Read page from disk into buffer pool  
(if not already present)

- ▶ **Flush (page)**

Evict page from buffer pool  
and write to disk

- ▶ **Release (page)**

Evict page from buffer pool  
without writing to disk

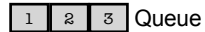


# Buffer Replacement Policy

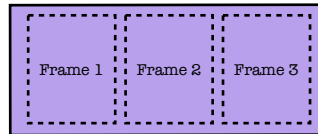
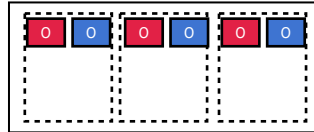
- ▶ Which page to evict from the buffer pool?
- ▶ Policies
  - **L**east-**R**ecently **U**sed (LRU)
  - **M**ost-**R**ecently **U**sed (MRU)
  - Clock
  - First-in First-out (FIFO)
  - Random
  - ...
- ▶ Choice of policy impacts speed, accuracy, and correctness

# LRU

## Example



Page Table

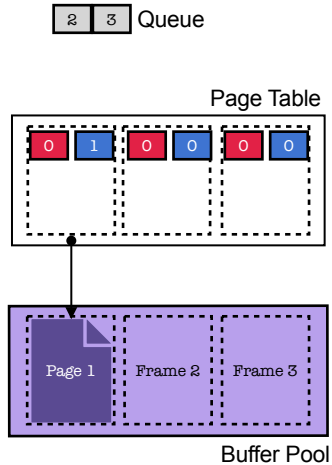


Buffer Pool

# LRU

## Example

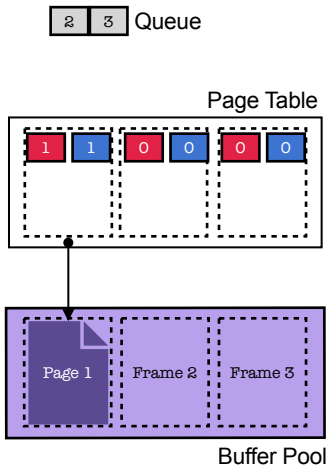
1. request Page 1



# LRU

## Example

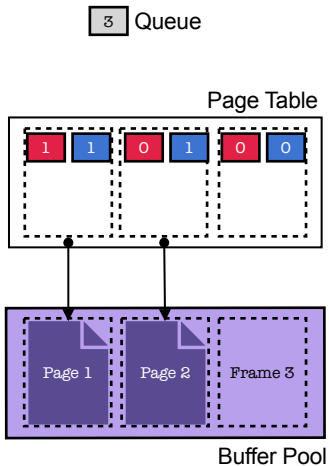
1. request Page 1
2. modify Page 1



# LRU

## Example

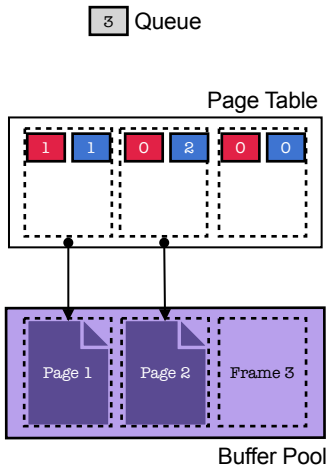
1. request Page 1
2. modify Page 1
3. request Page 2



# LRU

## Example

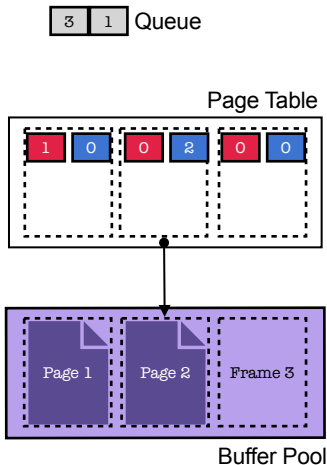
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2



# LRU

## Example

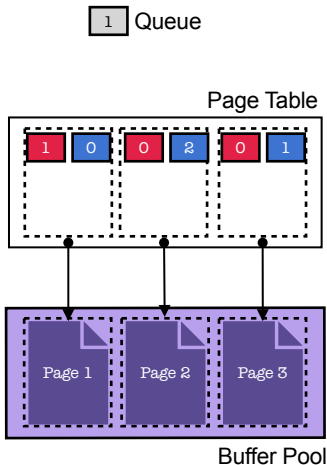
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1



# LRU

## Example

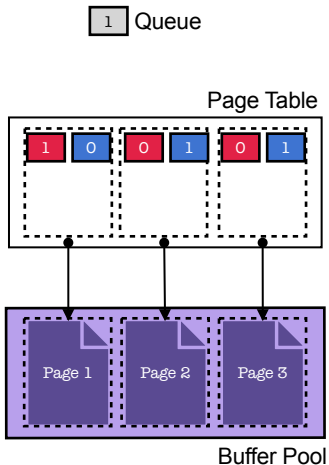
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3



# LRU

## Example

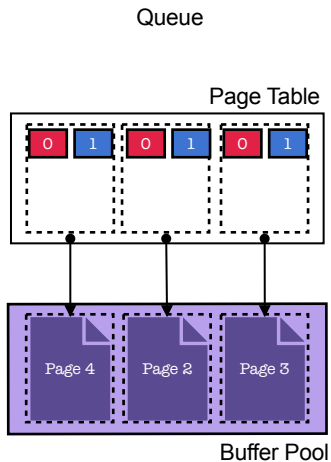
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2



# LRU

## Example

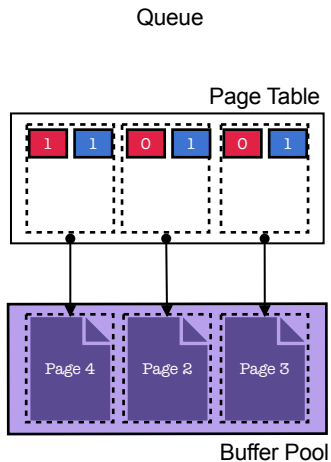
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4



# LRU

## Example

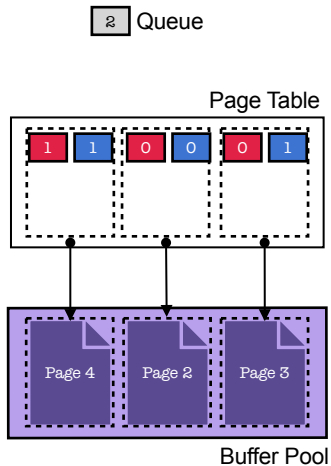
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4



# LRU

## Example

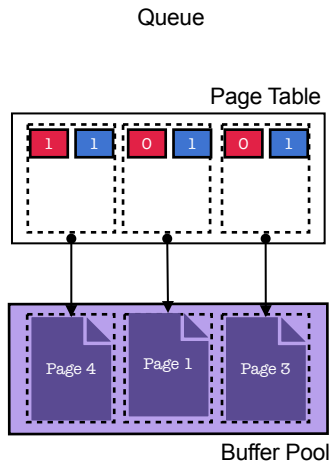
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2



# LRU

## Example

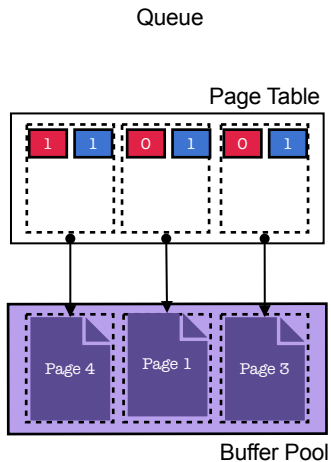
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2
11. request Page 1



# LRU

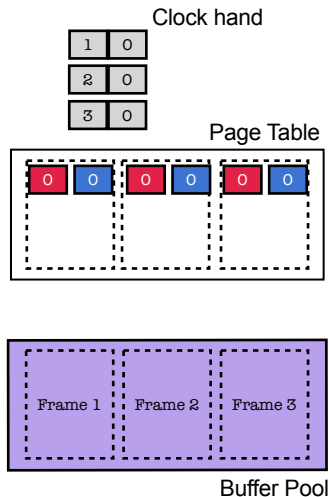
## Example

1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2
11. request Page 1
12. request Page 5



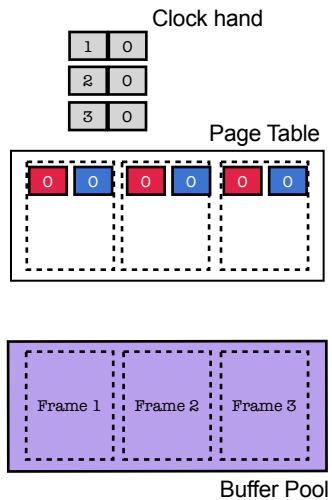
# CLOCK

- ▶ LRU variant with lower memory
- ▶ Frames as organized as circular buffer
- ▶ Maintains a reference bit for each frame
  - Set to 1 when pin count = 0
- ▶ Pointer to frame (clock hand)
- ▶ if pin count  $> 0$ , increment pointer
- ▶ if ref. bit = 1, reset and increment pointer
- ▶ if ref. bit = 0, and pin count = 0, select the page



# CLOCK

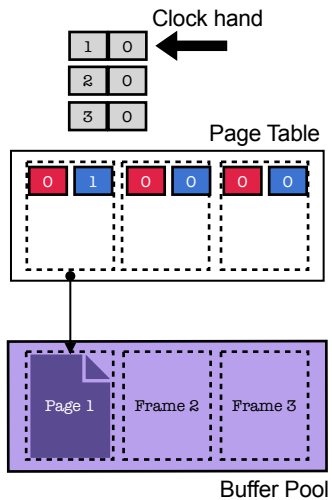
## Example



# CLOCK

## Example

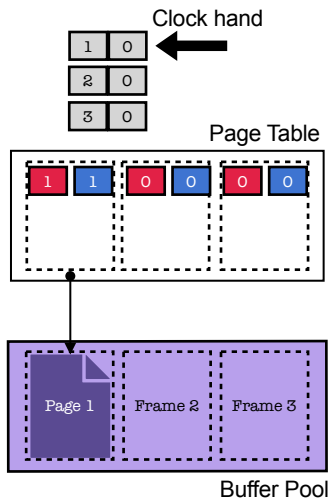
1. request Page 1



# CLOCK

## Example

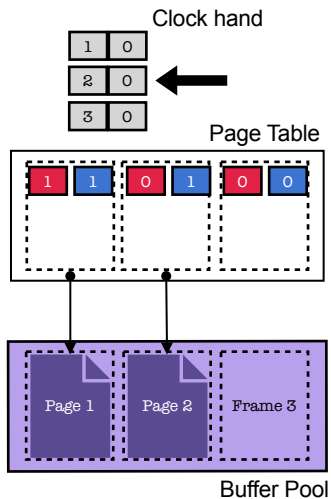
1. request Page 1
2. modify Page 1



# CLOCK

## Example

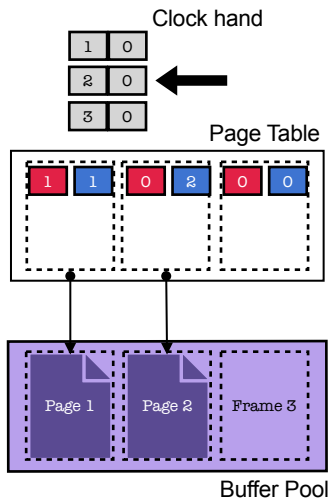
1. request Page 1
2. modify Page 1
3. request Page 2



# CLOCK

## Example

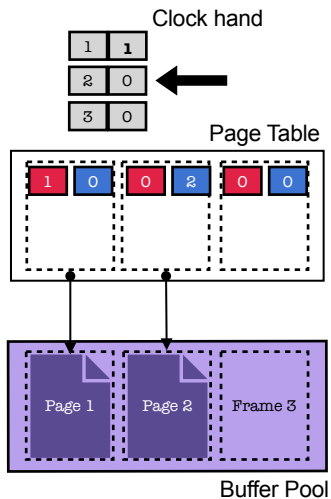
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2



# CLOCK

## Example

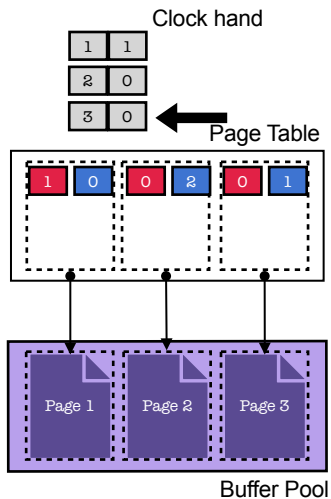
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1



# CLOCK

## Example

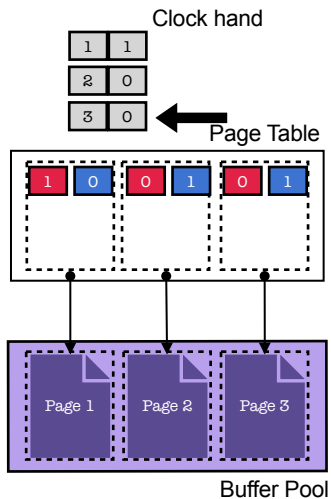
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3



# CLOCK

## Example

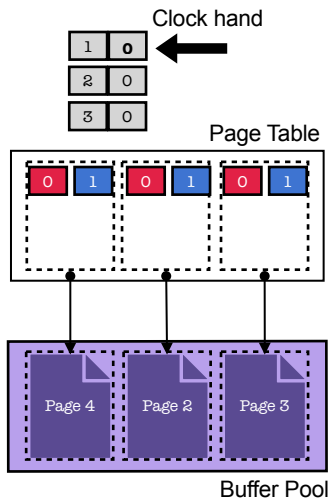
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2



# CLOCK

## Example

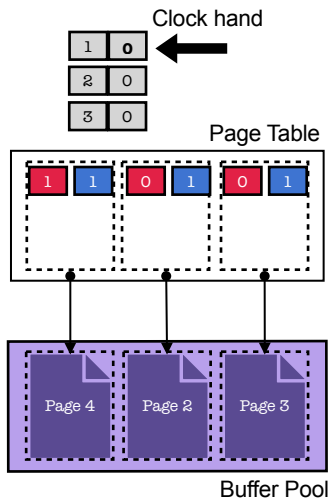
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4



# CLOCK

## Example

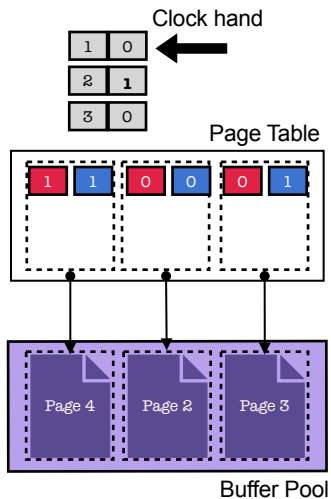
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4



# CLOCK

## Example

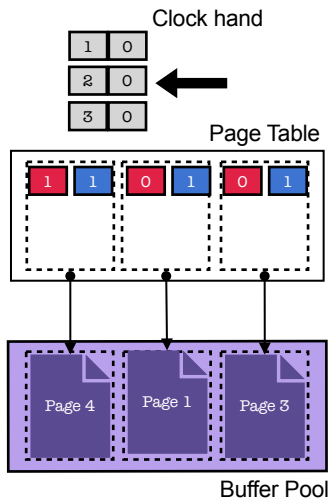
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2



# CLOCK

## Example

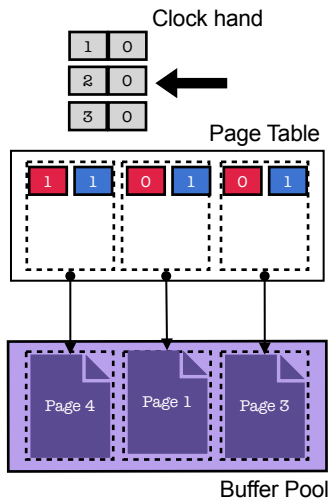
1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2
11. request Page 1



# CLOCK

## Example

1. request Page 1
2. modify Page 1
3. request Page 2
4. request Page 2
5. release Page 1
6. request Page 3
7. release Page 2
8. request Page 4
9. modify Page 4
10. release Page 2
11. request Page 1
12. request Page 5



# Sequential Flooding

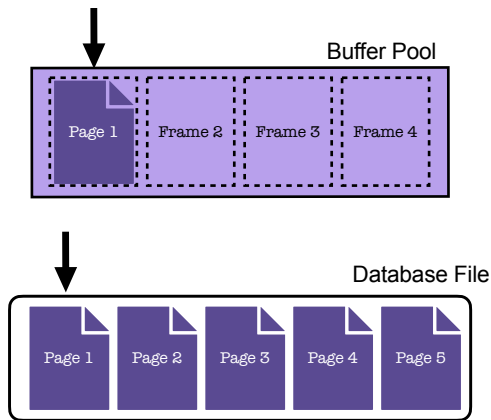
- ▶ `select * from table where table.col = val`
  - Sequential Scan
- ▶ LRU and CLOCK can lead to **sequential flooding**
- ▶ Pages are only read once, and not used again
- ▶ MRU is preferable over LRU for repeated sequential scans

# Optimizations

## ► Pre-fetching pages

e.g., based on the query plan ('ll be covered in later lectures)

- **Sequential Scans**
- Index Scans

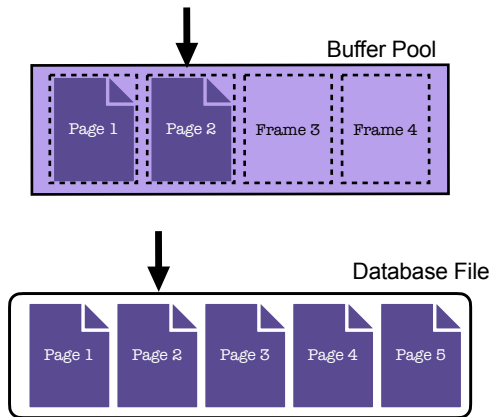


# Optimizations

## ► Pre-fetching pages

e.g., based on the query plan ('ll be covered in later lectures)

- **Sequential Scans**
- Index Scans

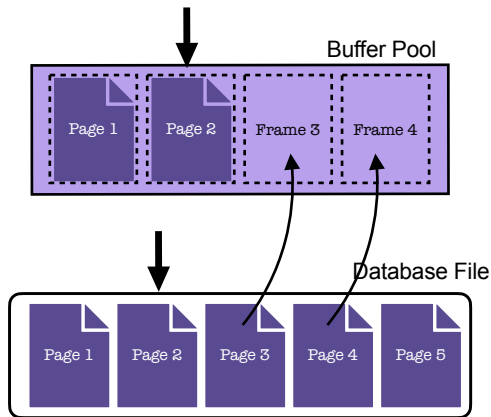


# Optimizations

## ► Pre-fetching pages

e.g., based on the query plan ('ll be covered in later lectures)

- **Sequential Scans**
- Index Scans

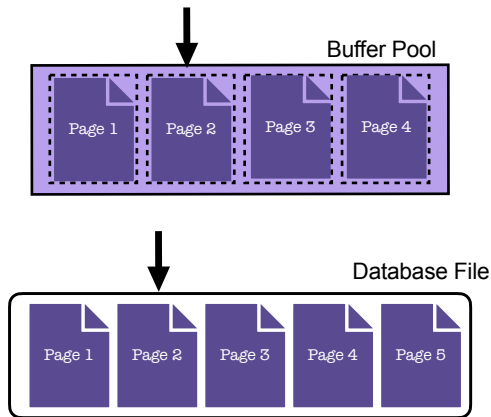


# Optimizations

## ► Pre-fetching pages

e.g., based on the query plan ('ll be covered in later lectures)

- **Sequential Scans**
- Index Scans

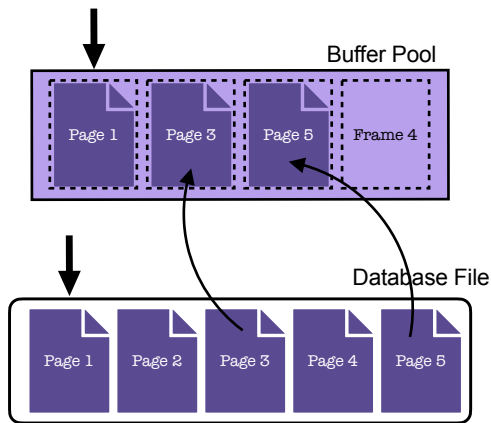
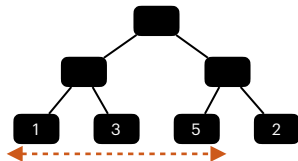


# Optimizations

## ► Pre-fetching pages

e.g., based on the query plan ('ll be covered in later lectures)

- Sequential Scans
- **Index Scans**



# Other Aspects

- ▶ Optimization
  - Scan sharing
- ▶ Bypassing OS page cache
- ▶ DBMS may have more than one buffer pool
  - Per page
  - Per database
  - Multiple instances
- ▶ Buffer pool management
  - Global policies — Decisions based on all running queries
  - Local policies — Decisions only based on specific query